METRIC

A-A-54687
22 June 1992

COMMERCIAL ITEM DESCRIPTION

CARTRIDGE, WATER DEMINERALIZER

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-D-37524 which is cancelled.

This Commercial Item Description covers the requirements for an ion exchange resin cartridge for use with a hand-operated water demineralizer.

Salient characteristics:

Shall be an ion exchange resin cartridge containing a mixture of cation and anion exchange resins. The cation resins shall convert salts (e.g. sodium chloride) dissolved in water to their corresponding acids. The anion resins shall absorb the acids produced by the cation resins.

The cartridge shall be fabricated of plastic and contain ion exchange resins and an indicator. The sides of the cartridge shall have sufficient clarity for checking the color of the indicator. The base of the cartridge shall have a protuberance with an outside diameter of 16.0 mm ± .13 mm. The cartridge shall be capable of being inserted into a 473 ml bottle having the following dimensions: 171 ±4 mm high and a 70 ±4 mm outside diameter. With a full bottle of water attached to the cartridge and squeezed the water shall pass through the cartridge and there shall be no evidence of leakage from where the bottle and cartridge are connected.

AMSC N/A FSC 4610

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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The cartridge shall contain enough resins to remove 2.21 grams of sodium chloride from a solution of sodium chloride and water. The ion exchange resins shall be a mixture of cation and anion resins. A blue indicator shall be applied to the anion resins prior to mixing them with the cation resins. The change in the color of the indicator shall be apparent as a yellowish brown band. The band shall start at the base of the cartridge and expand upward as the ion exchange capacity of the cartridge is expended. The cartridge shall have a permanent discard mark. The mark shall indicate that the cartridge is to be discarded when the yellowish brown band reaches the mark.

The initial use of the cartridge shall allow 6.6 liters of salt solution, containing 2.21 grams (35 grains) of sodium chloride, to pass through the cartridge and be filtered to produce water without resin particles and containing no more than 10 parts per million of sodium chloride.

Workmanship. The cartridge shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Package (PG). One package containing two cartridges, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspection and Testing. Inspection and testing shall be conducted on the water demineralizer cartridge in accordance with commercial practice. The sampling for inspection and testing shall be in accordance with commercial practice.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. (Intermediate/exterior package quantities, allowable expiration dating period, labeling and marking must be specified in the contract and/or order).

NOTE: The following National Stock Number is covered by this document:

NSN	Item Identification
4610-00-299-8480	CARTRIDGE, WATER DEMINERALIZER

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 4610-0128

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54687	2. DOCUMENT DATE (YYMMDD) 22 JUNE 1992

3. DOCUMENT TITLE
CARTRIDGE, WATER DEMINERALIZER

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Personnel Support Center ATTN: MSE	b. TELEPHONE (Include Area Code) (1) Commercial: (215)737-2870 (2) AUTOVON: 444-2870
c. ADDRESS (Include Zip Code) 2800 S. 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — Previous editions are obsolete. — 98-290

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